



Effect of temperature on the performance of electrochemical seismic sensor and the compensation method

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ABSTRACT

In this paper, the influence of temperature on the performance parameters of the electrochemical seismic sensor is studied. The sensitivity and temperature characteristics of the electrochemical seismic sensor at different temperatures are measured, and the effects of temperature on the sensitivity and amplitude-frequency characteristics of the electrochemical seismic sensor are analyzed. Based on the theoretical formula model of transfer function and experimental data, the temperature drift model of electrochemical seismic sensor is established, and the temperature compensation model of sensitivity is given. The sensitivity drift amplitude of the electrochemical seismic sensor measured decreases obviously after data correction with compensation formula, and the sensitivity change rate decreased from 45% to 7% in the frequency band.

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1. Research background and significance

Seismic exploration is widely used in the study of the earth's internal structure, monitoring of nuclear explosions, engineering exploration and detection, geological hazard prediction and so on [1–3]. With the increasing depth of seismic exploration, the high-frequency signal in seismic wave attenuates seriously in the transmission process, and the signal received by seismic sensor is mainly composed of low-frequency components [4,5]. Therefore, seismic sensor with low frequency signal detection ability plays an important role in deep seismic exploration, and its performance should be stable in practical application to ensure the reliability of the quality of data acquisition [6–8].

The electrochemical seismic sensor developed on the basis of molecular-electronic induction technology is a new type of seismic sensor, in recent years, it has been gradually valued and applied in the field of low-frequency signal detection [9–11]. Compared with traditional electromechanical seismic sensor, electrochemical seismic sensor has the advantages of small size, high sensitivity and good low-frequency characteristics [12,13]. But in practical application, compared with traditional electromechanical seismograph, the working temperature of electrochemical seismic sensor is limited by the boiling point and solidification point of electrolyte in

the electrochemical reaction chamber, and the solution viscosity and ion diffusion coefficient change exponentially with temperature, which results in that the performance parameters of the electrochemical seismic sensor are greatly affected by temperature [14–16]. The experimental environment of field seismic exploration is complex and the temperature difference between day and night is large. When the exploration precision is high, the influence of temperature change on the performance of the electrochemical seismic sensor cannot be ignored. It is of great significance for the practical application of the electrochemical seismic sensor to study the law of the output characteristics changing with temperature.

2. Effect of temperature on electrochemical seismic sensor

The working environment temperature of the electrochemical seismic sensor is controlled by the temperature control system, and the output of the electrochemical seismic sensor is tested by the ultra-low frequency vibration calibration system. The single component electrochemical seismic sensor MTL510 is taken as the test object. Before measurement, put the seismic sensor into the temperature control box of corresponding temperature for more than 5 h. The temperature of the sensor can be regarded as the set temperature. The experimental environment is shown in Fig. 1. The technical parameters of the ultra-low frequency vibration calibration system are shown in the Table 1.

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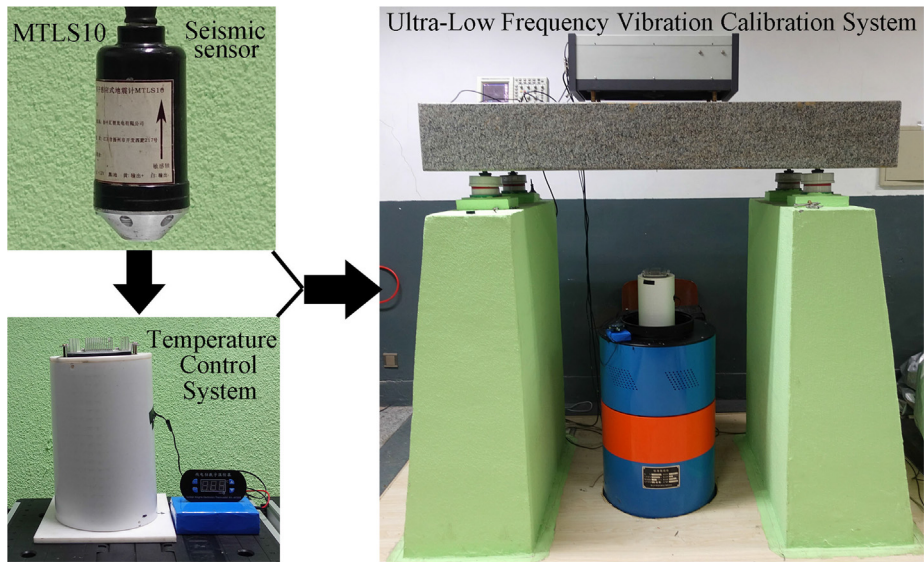


Fig. 1. Experimental test environment.

Table 1
Ultra-low frequency vibration calibration system technical parameters.

Name	Characteristic parameters of CDZ-100 vertical vibration table
Frequency Range	200 s-160 Hz
Maximum Displacement	100 mm(p-p)
Maximum Velocity (peak)	0.2 m/s
Full-load Maximum Acceleration (peak)	10 m/s ²
Distortion	Displacement Distortion ≤ 1%(f < 0.1 Hz) Acceleration Distortion ≤ 1%(f > 0.1 Hz)

Because the maximum working temperature recommended by the electrochemical seismic sensor is not more than 55°C, and according to the working environment temperature of the seismic sensor in practical application, five temperature points are selected for experimental testing. Fig. 2 shows the temperature dependence of the sensitivity of the electrochemical seismic sensor at 5 Hz.

From the curve of Fig. 2, it can be seen that the output sensitivity of the electrochemical seismic sensor varies greatly with temperature. With the increase of temperature, the output sensitivity

of the electrochemical seismic sensor increases, and from 10 °C to 45 °C, the sensitivity changes by more than 45%. From the change of curve slope in the figure, it can be seen that the sensitivity change rate of the electrochemical seismic sensor increases with the increase of temperature, and the influence of temperature change on the sensitivity of the electrochemical seismic sensor is non-linear.

The sensitivity-temperature characteristic curve of the whole frequency band of the electrochemical seismic sensor is shown in Fig. 3. From the curves in the figure, it can be seen that the sensitivity of electrochemical seismic sensor at each test frequency point in the whole frequency band increases with the increase of temperature, and from 10°C to 45°C, the sensitivity changes by more than 45%, which is the same as that of single frequency point. The analysis of the experimental results shows that the flatness, sensitivity and amplitude-frequency characteristics of the frequency band of the electrochemical seismic sensor will be affected when the temperature rises.

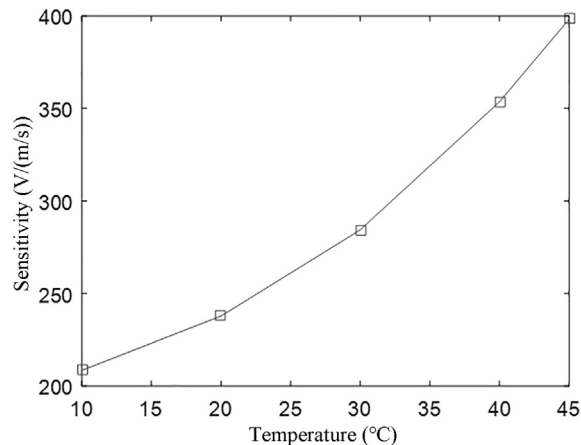


Fig. 2. Temperature-dependent sensitivity curve of electrochemical seismic sensor.

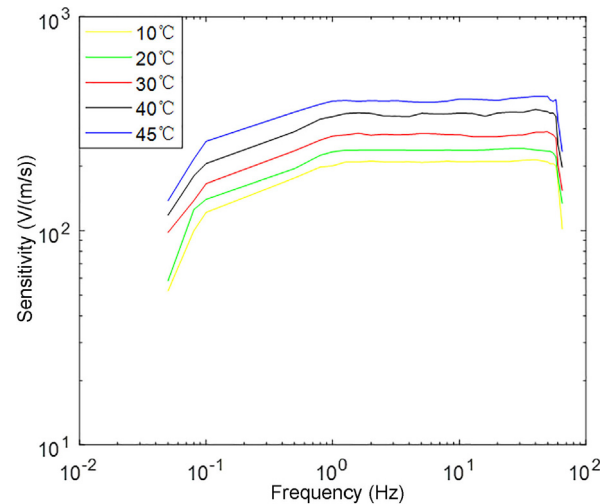


Fig. 3. Sensitivity temperature characteristic curve of electrochemical seismic sensor in full frequency band.

3. Compensation method for the effect of temperature on the sensitivity of electrochemical seismic sensor

The electrochemical seismic sensor uses liquid electrolyte as inertial mass to detect vibration signals. The electrochemical reaction chamber is filled with water solution mixed by liquid potassium iodide and iodine in a certain proportion. The sensitive core is located in the center of the electrochemical reaction chamber, and the electrolyte can flow freely in the micro-channel of the sensitive core. The two sides of the electrochemical reaction chamber are elastic membranes, which seal the electrolyte in the electrochemical reaction chamber and provide feedback force for the movement of the internal electrolyte. As shown in Fig. 4, when the electrochemical seismic sensor is well coupled with the ground and is accelerated in one direction to the left, the internal electrolyte is subjected to a right-facing inertial force. When the inertia force is greater than the flow resistance of the electrolyte in the electrochemical reaction chamber, the internal electrolyte will flow to the right along the channel direction of the electrochemical reaction chamber [10,11].

Let $u(t)$ be the ground displacement along the flow path of the reaction chamber and $x(t)$ be the ground displacement of the electrolyte in the reaction chamber, both of which move to the right. The force analysis of the liquid inertia mass in the reaction chamber is carried out. Firstly, when the internal electrolyte flows to the right, it will be subject to the feedback force of the right elastic mode, the size of which is $-kV(t)$. $V(t)$ is the fluid volume flowing through the micro-channel of the sensitive core in the reaction chamber, k is the volume stiffness coefficient, which only depends on the nature of the elastic film itself. The negative feedback force is because the feedback force prevents the internal electrolyte from flowing to the right. The flow direction is opposite to that of the internal electrolyte. The second force is the damping force, whose magnitude is proportional to the fluid impedance R_h of the internal electrolyte. The damping force is negative in the opposite direction of the electrolyte flow. The magnitude can be calculated by formula (3.1) [15–21].

$$F_d = -R_h S_{ch} dV(t)/dt \quad (3.1)$$

Among them, S_{ch} is the cross-sectional area of the flow passage of the sensitive core in the reaction chamber.

According to Newton's second law of classical mechanics, the equation of motion of the liquid inertia mass in the reaction chamber of the electrochemical seismic sensor is as follows:

$$-R_h S_{ch} \frac{dV(t)}{dt} - kV(t) = m \frac{d^2 x(t)}{dt^2} + ma(t) \quad (3.2)$$

The correlation can be calculated by the following formula:

$$V(t) = S_{ch} x(t) \quad (3.3)$$

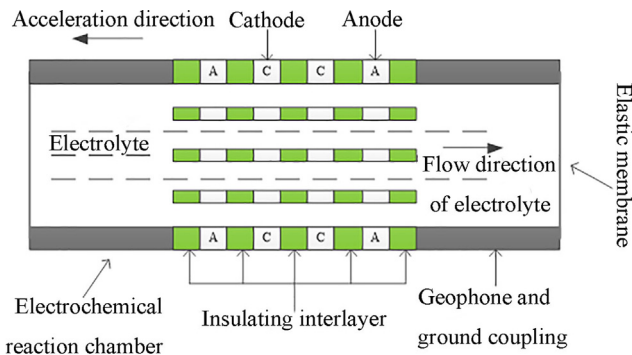


Fig. 4. Schematic diagram of the structure of electrochemical reaction chamber.

$$m = \rho L S_{ch} \quad (3.4)$$

$$a(t) = \frac{d^2 u(t)}{dt^2} \quad (3.5)$$

In the formula, m is the mass of the internal electrolyte, ρ is the density of the electrolyte, and L is the length of the runner of the sensitive core in the reaction chamber.

From the relation $Q(t) = dV(t)/dt$, $Q(t)$ is volume flow, and the equation (3.2) is converted from time domain to frequency domain. The mechanical part of the transfer function of electrochemical seismic sensor is obtained as follows:

$$|H_{mech}(w)| = \frac{|Q(w)|}{|a(w)|} = \frac{\rho L}{\sqrt{\left(\frac{\rho L}{S_{ch}}\right)^2 \frac{(w^2 - w_0^2)^2}{w^2} + R_h^2}} \quad (3.6)$$

Type $w_0 = \sqrt{k/\rho L}$ is the mechanical resonance frequency of electrochemical seismic sensor.

When the electrochemical seismic sensor works normally, the effective ions in the reaction chamber will undergo redox reaction at the electrode of the sensitive core. Without the excitation of external vibration signal, the ion concentration in the solution is in equilibrium, and the electrochemical reaction rate at the two cathodes is the same, so the output differential current is zero. When the electrochemical seismic sensor vibrates with the ground, under the excitation of external vibration signals, the internal electrolyte flows, resulting in the breakdown of the ion concentration balance between the electrodes of the sensitive core, and the differential current output at the two cathodes is not zero. The electrochemical part of the transfer function of the electrochemical seismic sensor converts the flow information of the internal electrolyte into the output of the electric signal, which can be approximately expressed by the following formula:

$$|H_{ec}(w)| \sim \frac{1}{\sqrt{1 + \left(\frac{w}{w_D}\right)^2}} \quad (3.7)$$

In the formula, $W_D = D/d^2$ is the diffusion frequency, D is the diffusion coefficient of the effective ions in the electrolyte, and d is the distance between the cathode and the anode.

The electrochemical seismic sensor first converts the external vibration information into the flow of electrolyte in the reaction chamber, and then converts the sensitive core into the output of the electric signal. The theoretical form of the transfer function of the electrochemical seismic sensor in frequency domain is obtained by combining (3.6) and (3.7):

$$|H(w)| \sim \frac{\rho L}{\sqrt{\left(\frac{\rho L}{S_{ch}}\right)^2 \frac{(w^2 - w_0^2)^2}{w^2} + R_h^2}} \frac{1}{\sqrt{1 + \left(\frac{w}{w_D}\right)^2}} \quad (3.8)$$

Referring to the theoretical formula model of the transfer function of the electrochemical seismic sensor and fitting the sensitivity curve of the electrochemical seismic sensor based on the experimental data, the temperature drift model of the sensitivity of the electrochemical seismic sensor is established as shown in equation (3.9).

$$|F(w)| = \frac{A_0 w}{\left(1 + \frac{w^2}{w_1^2}\right)^{0.5} \left(1 + \frac{w^2}{w_2^2}\right)^{0.5}} \quad (3.9)$$

w_1 , w_2 , A_0 is a temperature-dependent coefficient. Among them, A_0 is the correlation coefficient of the influence of temperature on the sensitivity amplitude of the electrochemical seismic sensor, and w_1 , w_2 is the correlation coefficient of the frequency turning point of the electrochemical seismic sensor.

The coefficient A_0 , which affects the sensitivity of electrochemical seismic sensor, is taken as the research object. The correlation between coefficient A_0 and temperature T is found by approximate fitting of the temperature drift model and the actual data. The approximate fitting effect between the temperature drift model and the sensitivity curve of the electrochemical seismic sensor is shown in Fig. 5(a)–(d). The results show that the temperature drift model of the electrochemical seismic sensor fits the measured data well in the pass band range of the electrochemical seismic sensor, and the fitting effect is poor in the high frequency part whose frequency band is higher than 50 Hz and the low frequency part whose frequency band is lower than 0.1 Hz. This is due to the limitation of the bandpass filter in the external conditioning circuit of the electrochemical seismic sensor, so the frequency is comparable. The sensitivity curve of seismic sensor decreases rapidly at high and low levels, which leads to larger fitting error.

Although the direct linear relationship between the amplitude coefficient A_0 and temperature T is poor, the experimental results show that the linear relationship between the logarithm of coefficient A_0 and temperature T is better. Fig. 6 shows the results of linear fitting of the relationship curve between the amplitude coefficient A_0 and temperature T by using the least square method. The fitting formulas are as follows:

$$\lg(A_0) = 0.0076T + 2.9 \quad (3.10)$$

The sensitivity of seismic sensor is invariable after testing and calibration, and the output sensitivity of the frequency band of seismic sensor communication is usually taken as the temperature T in formula (3.10), the compensation formula of the output sensitivity of the electrochemical seismic sensor measured with the change of temperature can be further established as follows:

$$S_{TC} = \frac{A_0(20) \cdot S(T)}{A_0(T)} \quad (3.11)$$

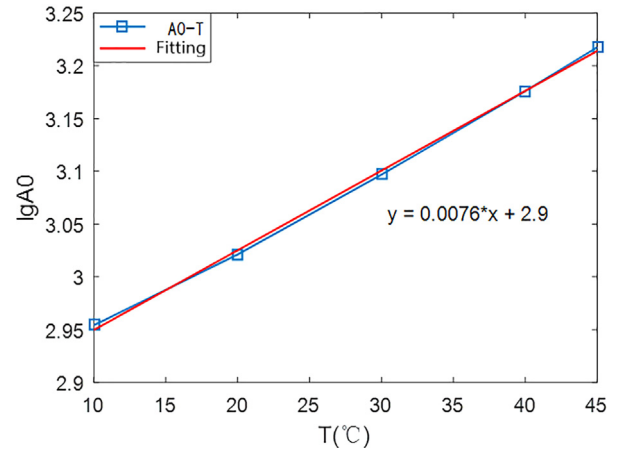


Fig. 6. Curve of the relationship between A_0 parameters and temperature T .

In the formula, $S(T)$ is the output sensitivity of the electrochemical seismic sensor when the ambient temperature is T , S_{TC} is the output sensitivity of the electrochemical seismic sensor compensated by data correction, and $A_0(T)$ is the value of coefficient A_0 when the temperature is T .

The output sensitivity curve of the electrochemical seismic sensor measured at different temperatures can be corrected according to equation (3.11). The sensitivity curve of the electrochemical seismic sensor measured after compensation is compared with that before compensation as shown in Fig. 7(a)–(b). From the comparison of the above two graphs, it can be seen that the output sensitivity curves of the electrochemical seismic sensor measured at different temperatures are significantly reduced after the correction of the compensation formula.

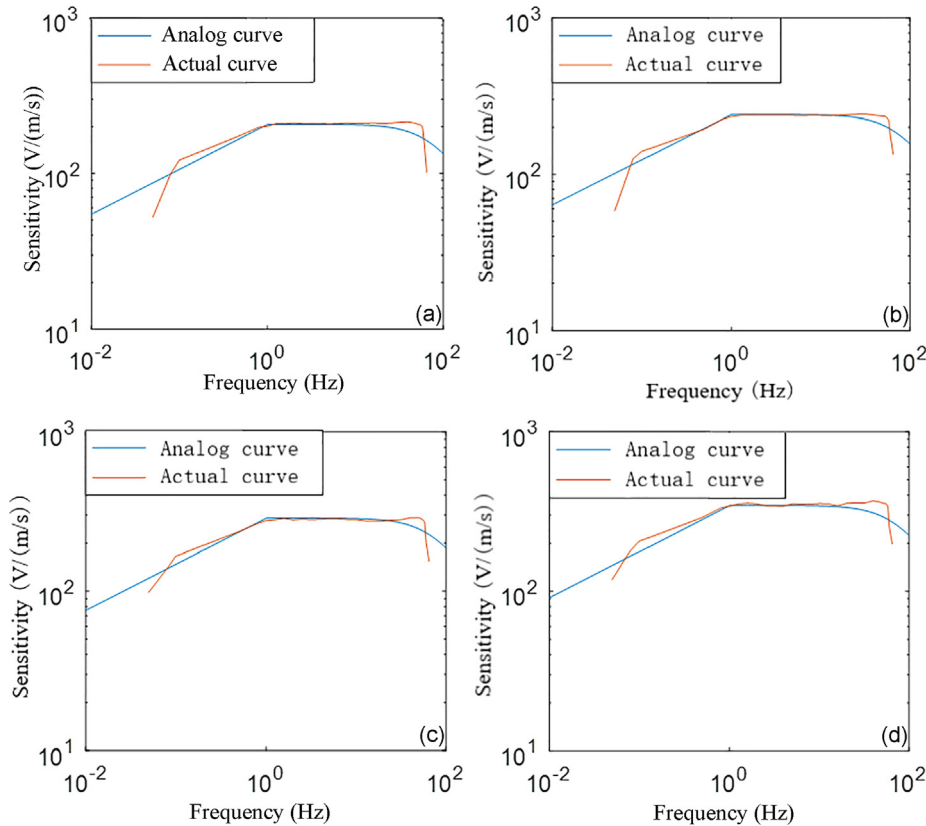


Fig. 5. (a)–(d) Temperature drift model and fitting curve of actual data at different temperatures, (a): 10 °C, (b): 20 °C, (c): 30 °C, (d): 40 °C.

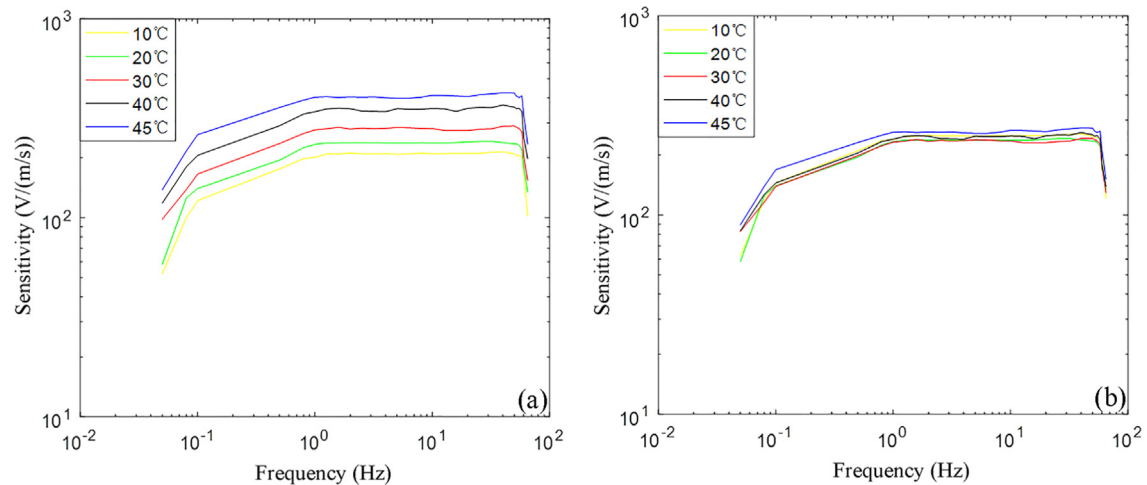


Fig. 7. (a)–(b) Sensitivity curve of electrochemical seismic sensor measured before compensation and after compensation, (a): before, (b): after.

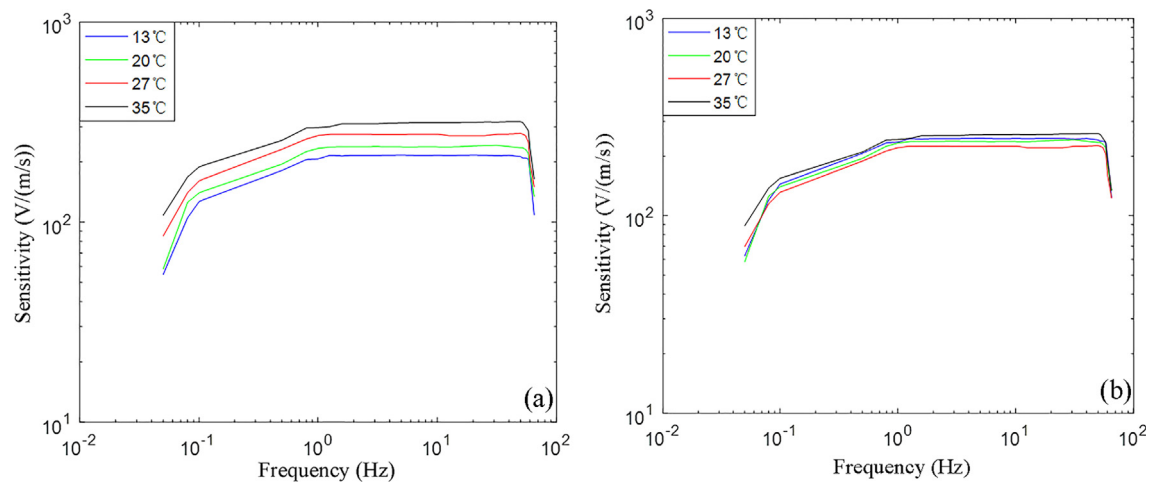


Fig. 8. (a)–(b) Sensitivity curve of electrochemical seismic sensor before compensation and after compensation, (a): before, (b): after.

In order to further illustrate the validity of the temperature drift model and compensation method established in this paper, the sensitivity curves of electrochemical seismic sensor at different temperatures are tested by the same test method, and the output data of electrochemical seismic sensor are corrected by the temperature drift model and compensation formula established in this paper. Fig. 8(a)–(b) show the sensitivity curves of the electrochemical seismic sensor before and after compensation at different temperatures. It can be seen from the figure that the temperature drift amplitude of the electrochemical seismic sensor is obviously reduced after compensation. Comparing with the sensitivity of the passband at 20 °C, the sensitivity of the electrochemical seismic sensor is compensated, and the change rate of the sensitivity of the passband at different temperatures is not more than 7%.

The model is verified by the electrochemical seismic sensor MTSS-1001 (1 Hz–50 Hz), and the sensitivity change after compensation is also obviously reduced. According to the experimental results, the temperature drift model can be used to compensate the output sensitivity of different types of electrochemical seismic sensors.

4. Summary

This paper proposed and demonstrated the temperature drift model of the electrochemical seismic sensor. The temperature drift

amplitude of the electrochemical seismic sensor is obviously reduced after data correction, and the sensitivity change rate decreased from 45% to 7% in the frequency band. It is verified by the electrochemical seismic sensor MTSS-1001, the results show that the model is suitable for other electrochemical seismic sensors.

CRediT authorship contribution statement

Jun Lin: Project administration, Supervision. **Huan Gao:** Writing - original draft, Writing - review & editing. **Xiufeng Wang:** Data curation, Formal analysis, Funding acquisition, Writing - original draft. **Chunyan Yang:** Funding acquisition. **Yi Xin:** Project administration. **Xianfeng Zhou:** Funding acquisition, Project administration, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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